P.4 SCIENCE

LESSON PLANS

TERM 1 2026

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PRIMARY FOUR SCIENCE LESSON PLANS TERM ONE 2026 -BY MY. JAYSON

LESSON 1: PLANT LIFE - INTRODUCTION TO PLANTS

DATE	CLASS	SUBJECT / L. AREA	NO. OF CHILDREN	TIME
	Primary Four	Integrated Science		60 Minutes

THEME	TOPIC	CONTENT
The World of Living Things	Plant Life	Introduction to Plants

COMPETENCES:

- **Subject Competence:** Learners identify plants as living things that make their own food and classify them into flowering and non-flowering groups.
- **Language Competence:** Learners pronounce and use scientific vocabulary such as chlorophyll, flowering, non-flowering, and spores correctly.

TEACHING METHODS/TECHNIQUES:

- Guided discovery
- Question and answer
- Group discussion
- Observation
- Explanation
- Brainstorming
- Demonstration

LEARNING OUTCOMES:

By the end of this lesson, learners should be able to:

1. Define what a plant is.
2. State the characteristics of plants.
3. Identify the two main groups of plants.
4. Give examples of flowering and non-flowering plants.

TEACHING/LEARNING MATERIALS:

- Real plant specimens (small potted plant, moss, fern)
- Charts showing groups of plants

- Pictures of flowering and non-flowering plants
- Chalkboard/whiteboard and markers
- P.4 SUREKEY SCIENCE NOTES (Pages 6-7)
- Diagram of non-flowering plants (Page 6)

LESSON DEVELOPMENT

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	**ENGAGE**	- Asks learners to name plants around the school compound - Shows a real plant and asks what it is - Asks learners: "What do plants need to live?"	- Brainstorming - Question and answer	- Mentions names of plants seen around school - Identifies the object as a plant - Suggests things plants need (water, sunlight, soil)
15 mins	**EXPLORE**	- Divides learners into small groups - Gives each group pictures of different plants - Asks groups to sort plants into two groups based on what they observe - Moves around guiding groups	- Group work - Guided discovery - Observation	- Observes pictures carefully - Discusses with group members - Sorts plants into groups - Shares ideas with other groups
20 mins	**EXPLAIN**	- Defines plants: "Plants are organisms that make their own food through photosynthesis and are typically rooted in the ground." (Page 6) - Lists characteristics of plants: • They make their own food • They have many cells • They contain chlorophyll • They are rooted in the ground - Introduces two groups of	- Explanation - Lecture - Demonstration	- Listens attentively - Writes notes in books - Repeats key terms after teacher - Observes the diagram on Page 6 - Names non-flowering plants from the diagram

		plants: flowering and non-flowering - Defines non-flowering plants: plants that do not bear flowers (Page 6) - Mentions examples: mosses, liverworts, ferns, pine, fir - Refers learners to the diagram on Page 6		
10 mins	**ELABORATE**	- Asks learners to identify flowering plants around the classroom - Gives examples of flowering plants from Page 6: maize, beans, banana, mango, tomato - Asks learners to state whether a given plant is flowering or non-flowering - Guides learners to complete Evaluation activity 1.1 questions 1-4 (Page 8)	- Question and answer - Guided practice	- Names flowering plants seen around - Writes examples of flowering plants - Identifies plants as flowering or non-flowering - Attempts questions 1-4 with teacher's guidance
5 mins	**EVALUATE**	- Asks oral questions to check understanding: • What is a plant? • Name two groups of plants • Give one example of a non-flowering plant - Gives class work from the source material	- Oral questioning - Written exercise	- Answers oral questions - Writes class work in books
5 mins	**EXTEND**	- Asks learners to collect and bring two different leaves (one from a flowering plant and one from a non-flowering plant) for next lesson - Tells learners to read about dicotyledonous and monocotyledonous plants on Pages 7-8	- Giving assignment - Setting homework	- Listens to home assignment - Collects leaves as instructed

CLASS WORK:

From Evaluation activity 1.1 (Page 8)

1. Name the process that enables plants to make their own food.
2. Mention two groups of plants.
3. State any two characteristics of plants.
4. Give the reason why a carrot plant is grouped under flowering plant.

LESSON EVALUATION

STRENGTHS:	AREAS WHICH NEED IMPROVEMENT:	WAY FORWARD:

LESSON 2: PLANT LIFE - FLOWERING PLANTS (MONOCOTS AND DICOTS)

DATE	CLASS	SUBJECT / L. AREA	NO. OF CHILDREN	TIME
	Primary Four	Integrated Science		60 Minutes

THEME	TOPIC	CONTENT
The World of Living Things	Plant Life	Flowering Plants - Monocotyledons and Dicotyledons

COMPETENCES:

- **Subject Competence:** Learners differentiate between monocotyledonous and dicotyledonous plants based on their characteristics.
- **Language Competence:** Learners correctly pronounce and use terms such as monocotyledon, dicotyledon, venation, taproot, and fibrous root.

TEACHING METHODS/TECHNIQUES:

- Observation
- Comparison
- Group discussion
- Question and answer
- Demonstration
- Explanation

- Classification

LEARNING OUTCOMES:

By the end of this lesson, learners should be able to:

1. Define monocotyledonous and dicotyledonous plants.
2. State the characteristics of monocotyledonous plants.
3. State the characteristics of dicotyledonous plants.
4. Give examples of monocot and dicot plants.

TEACHING/LEARNING MATERIALS:

- Real specimens: bean seed, maize grain, onion bulb
- Leaves with parallel venation (maize, grass)
- Leaves with network venation (bean, mango)
- Charts showing monocot and dicot plants
- P.4 SUREKEY SCIENCE NOTES (Pages 7-8)
- Diagrams of monocot and dicot plants (Page 7)
- Leaves collected by learners

LESSON DEVELOPMENT

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	**ENGAGE**	- Asks learners to show the leaves they collected - Asks: "Are all leaves the same?" - Shows a bean leaf and a maize leaf and asks learners to observe the veins - Asks: "What do you notice about the veins on these leaves?"	- Observation - Question and answer	- Shows collected leaves - Observes differences in leaves - Compares bean leaf and maize leaf - Notices that veins are different
15 mins	**EXPLORE**	- Divides learners into groups - Gives each group different leaves and seeds - Asks groups to observe	- Group work - Guided discovery - Observation	- Observes leaves and seeds in groups - Sorts leaves into two groups (parallel and network veins)

		and sort leaves into two groups based on vein patterns - Asks groups to observe seeds (bean and maize) and note differences - Guides groups to record observations		- Discusses differences observed - Records findings
20 mins	**EXPLAIN**	- Introduces monocotyledonous plants: flowering plants with one cotyledon (Page 8) - States characteristics of monocots (Page 8): • Bear seeds with one cotyledon • Have parallel leaf venation • Store food in endosperm • Undergo hypogeal germination • Have fibrous root system - Introduces dicotyledonous plants: flowering plants with two cotyledons (Page 7) - States characteristics of dicots (Page 7): • Bear seeds with two cotyledons • Have network leaf venation • Store food in cotyledons • Undergo epigeal germination • Have taproot system - Refers learners to the table and diagram on Page 7 - Gives examples of monocots: maize, millet, sorghum, onion (Page 8) - Gives examples of dicots: beans, ground	- Explanation - Demonstration - Lecture	- Listens attentively - Takes notes - Observes diagrams on Page 7 - Names examples of monocots and dicots - Compares characteristics of both groups

		nuts, mango, orange (Page 7)		
10 mins	**ELABORATE**	- Shows different plants and asks learners to identify them as monocots or dicots - Asks: "To which group do cowpeas belong? Millet?" (Evaluation 1.1, Page 8) - Guides learners to complete Evaluation activity 1.1 questions 5-10 (Page 8) - Helps learners differentiate based on leaf venation and root systems	- Question and answer - Guided practice - Classification	- Identifies plants as monocots or dicots - Answers questions about cowpeas and millet - Completes Evaluation questions with guidance - Classifies plants based on observed features
5 mins	**EVALUATE**	- Asks oral questions: • What are monocotyledonous plants? • Give two characteristics of dicots • Name one monocot plant - Gives class work from the source material	- Oral questioning - Written exercise	- Answers oral questions - Writes class work in books
5 mins	**EXTEND**	- Asks learners to collect a taproot and a fibrous root from home - Tells learners to read about parts of a flowering plant on Pages 9-10	- Giving assignment - Setting homework	- Listens to home assignment - Prepares to collect roots

CLASS WORK:

From Evaluation activity 1.1 (Page 8)

5. Name the group of flowering plants to which;
   i) Cowpeas belong
   ii) Millet belongs
6. State any one difference between a moss plant and a bean plant.
7. Mention any one way in which plants are important to people.
8. How can one identify a monocotyledonous plant from other plants?

9. State the way in which a tomato plant reproduces.
10. Which group of flowering plants undergo epigeal germination?

LESSON EVALUATION

STRENGTHS:	AREAS WHICH NEED IMPROVEMENT:	WAY FORWARD:

LESSON 3: PLANT LIFE - PARTS OF A FLOWERING PLANT

DATE	CLASS	SUBJECT / L. AREA	NO. OF CHILDREN	TIME
	Primary Four	Integrated Science		60 Minutes

THEME	TOPIC	CONTENT
The World of Living Things	Plant Life	Parts of a Flowering Plant and Their Functions

COMPETENCES:

- **Subject Competence:** Learners identify the parts of a flowering plant and explain the functions of each part.
- **Language Competence:** Learners correctly use vocabulary such as terminal bud, axillary bud, shoot system, root system, and transport.

TEACHING METHODS/TECHNIQUES:

- Demonstration
- Observation
- Explanation
- Question and answer
- Diagram interpretation
- Group discussion
- Labeling

LEARNING OUTCOMES:

By the end of this lesson, learners should be able to:

1. Identify the main parts of a flowering plant.
2. State the functions of the stem, leaves, roots, and flowers.

3. Differentiate between the shoot system and root system.
4. Label the parts of a flowering plant on a diagram.

TEACHING/LEARNING MATERIALS:

- A complete potted plant (with roots, stem, leaves, flowers/fruits if possible)
- Chart showing parts of a flowering plant
- Diagrams from P.4 SUREKEY SCIENCE NOTES (Pages 9-10)
- Real specimens of roots, stems, leaves, flowers
- Manila paper and markers

LESSON DEVELOPMENT

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	**ENGAGE**	- Shows a complete potted plant to learners - Asks: "What are the visible parts of this plant?" - Lists learners' responses on the board - Asks: "What do you think each part does for the plant?"	- Demonstration - Brainstorming - Question and answer	- Observes the plant carefully - Names visible parts (roots, stem, leaves, flowers) - Suggests functions of each part
15 mins	**EXPLORE**	- Divides learners into groups - Gives each group a plant specimen or diagram - Asks groups to identify and list all parts they can see - Asks groups to discuss what each part might do - Guides groups to share their findings	- Group work - Guided discovery - Discussion	- Observes plant or diagram in groups - Identifies and lists plant parts - Discusses possible functions of each part - Presents findings to class
20 mins	**EXPLAIN**	- Draws a labeled diagram of a flowering plant on the board (refer to Page 9) - Explains the two systems of a flowering plant (Page 9): • Shoot system: grows above ground (stem, leaves, flowers, fruits) • Root system: grows underground (roots) - Explains functions of each	- Explanation - Lecture - Demonstration	- Draws and labels the diagram in books - Listens to explanations - Takes notes on functions of plant parts - Observes the diagram on Page 9 - Repeats key functions after teacher

		part using the table on Page 9: • Terminal bud: increases height • Axillary bud: grows into side branch, leaf, or flower • Stem: holds plant parts; transports water, mineral salts, and food • Leaves: make food for the plant • Roots: absorb water and mineral salts • Flowers: reproduction; develop into fruits and seeds • Fruits: store seeds - Refers learners to the diagram on Page 9		
10 mins	**ELABORATE**	- Shows learners a plant and asks them to identify each part and state its function - Asks: "What would happen to a plant if it had no roots?" "No leaves?" - Guides learners to complete Evaluation activity 1.2 questions 1-5 (Page 10) - Helps learners complete the table in question 9 (Page 10)	- Question and answer - Guided practice - Critical thinking	- Identifies plant parts and states functions - Answers questions about importance of parts - Completes Evaluation questions with guidance - Fills in the table correctly
5 mins	**EVALUATE**	- Asks oral questions: • Name the part that grows into a side branch • What is the main function of leaves? • Which system absorbs water from soil? - Gives class work from the source material	- Oral questioning - Written exercise	- Answers oral questions - Writes class work in books
5 mins	**EXTEND**	- Asks learners to draw and label a flowering plant at home - Tells learners to read about plant roots on Pages 10-11 - Asks learners to bring different types of roots for next lesson	- Giving assignment - Setting homework	- Listens to home assignment - Prepares to draw and label a plant

CLASS WORK:

From Evaluation activity 1.2 (Page 10)

1. Name the part of a flowering plant which grows into a side branch.
2. State the main function of leaves to the plant.
3. Identify the system of flowering plants that enables the plant to absorb water from the soil.
4. Name the part of the flowering plant that stores seeds.
5. Give any one function of the flower to an orange plant.

LESSON EVALUATION

STRENGTHS:	AREAS WHICH NEED IMPROVEMENT:	WAY FORWARD:

LESSON 4: PLANT LIFE - PLANT ROOTS

DATE	CLASS	SUBJECT / L. AREA	NO. OF CHILDREN	TIME
	Primary Four	Integrated Science		60 Minutes

THEME	TOPIC	CONTENT
The World of Living Things	Plant Life	Plant Roots - Taproot and Fibrous Root Systems

COMPETENCES:

- **Subject Competence:** Learners identify different types of root systems and explain the functions of roots and their parts.
- **Language Competence:** Learners correctly use terms such as radicle, osmosis, taproot, fibrous root, root hairs, and root cap.

TEACHING METHODS/TECHNIQUES:

- Demonstration
- Observation
- Comparison
- Explanation
- Question and answer
- Group discussion

- Diagram interpretation

LEARNING OUTCOMES:

By the end of this lesson, learners should be able to:

1. Define roots and state their functions.
2. Identify the parts of a root system.
3. Differentiate between taproot and fibrous root systems.
4. Give examples of plants with each root system.

TEACHING/LEARNING MATERIALS:

- Real root specimens: carrot, cassava (taproot); grass, maize (fibrous root)
- Charts showing root systems
- Diagrams from P.4 SUREKEY SCIENCE NOTES (Pages 11-12)
- Root specimens collected by learners
- Potted plants with visible roots

LESSON DEVELOPMENT

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	**ENGAGE**	- Shows a potted plant and gently pulls it out to expose roots - Asks: "What part of the plant is this?" - Asks: "Why do plants need roots?" - Shows different root specimens and asks learners to observe	- Demonstration - Question and answer	- Observes exposed roots - Identifies the part as roots - Suggests functions of roots - Observes different root specimens
15 mins	**EXPLORE**	- Divides learners into groups - Gives each group different root specimens - Asks groups to observe and sort roots into two groups - Asks groups to identify similarities and differences - Guides groups to draw what they observe	- Group work - Guided discovery - Observation	- Observes root specimens in groups - Sorts roots into two groups (taproot and fibrous) - Discusses differences between root types - Draws observed roots

20 mins	**EXPLAIN**	- Defines roots: "Roots are branching parts of a plant that grow downwards into the soil" (Page 10) - States that roots develop from the radicle (Page 10) - Explains functions of roots (Page 11): • Hold the plant firmly in soil • Absorb water and mineral salts through osmosis • Store food (root tubers like cassava, carrots) - Shows diagram of root system on Page 11 and explains parts: • Root hairs: absorb water and mineral salts • Root cap: protects the tip of growing root • Main root and lateral roots: hold plant firmly - Introduces root systems (Page 11): • Taproot system: main root with lateral roots (common in dicots like beans, mango, carrot) • Fibrous root system: all roots from same point, same size (common in monocots like maize, grass) - Refers learners to diagrams on Page 12	- Explanation - Lecture - Demonstration	- Listens attentively - Takes notes - Observes diagrams on Pages 11-12 - Identifies parts of root system - Names examples of plants with taproot and fibrous root
10 mins	**ELABORATE**	- Shows pictures of plants and asks learners to identify their root systems - Asks: "What type of root system does a carrot plant have? A maize plant?" (Evaluation 1.3, Page 12) - Guides learners to answer Evaluation activity 1.3 questions 1-5 (Page 12) - Helps learners label diagrams in question 8 (Page 12)	- Question and answer - Guided practice - Diagram interpretation	- Identifies root systems of given plants - Answers questions about carrot and maize - Completes Evaluation questions with guidance - Labels diagrams correctly

5 mins	**EVALUATE**	- Asks oral questions: • What part of the seed develops into roots? • Give two functions of roots • What is the function of root hairs? - Gives class work from the source material	- Oral questioning - Written exercise	- Answers oral questions - Writes class work in books
5 mins	**EXTEND**	- Asks learners to dig up a small plant and observe its root system - Tells learners to read about types of roots (adventitious roots) on Pages 13-15	- Giving assignment - Setting homework	- Listens to home assignment - Prepares to observe root systems

CLASS WORK:

From Evaluation activity 1.3 (Page 12)

1. Name the part of a seed that develops into the roots.
2. State any two functions of roots to a plant.
3. State the function of the following parts of the taproot system to a plant:
   (i) Root hair
   (ii) Root cap
4. Identify the difference in root structure between a carrot plant and a maize plant.
5. State the meaning of the term adventitious root.

LESSON EVALUATION

STRENGTHS:	**AREAS WHICH NEED IMPROVEMENT:**	**WAY FORWARD:**

LESSON 5: PLANT LIFE - TYPES OF ROOTS (ADVENTITIOUS ROOTS)

DATE	**CLASS**	**SUBJECT / L. AREA**	**NO. OF CHILDREN**	**TIME**
	Primary Four	Integrated Science		60 Minutes

THEME	TOPIC	CONTENT
The World of Living Things	Plant Life	Types of Roots - True Roots and Adventitious Roots

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